

B. Tech Degree IV Semester Examination April 2011

ME 406 MANUFACTURING PROCESS

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Define the process of successful casting, suitability of casting operation for a given material.
- (b) Explain the dry sand moulding.
- (c) Explain the process of casting cleaning.
- (d) Write notes on venting of moulds.
- (e) Write notes on hot rolling and cold rolling of metals.
- (f) Explain the process of surface treatment in metals and the types of surface coatings.
- (g) Explain the process of brazing and soldering and draw the Equilibrium phase diagram for Copper-silver alloy.
- (h) Explain the following welding process (i) Gas Welding (ii) Arc Welding

PART B

(4 x 15 = 60)

- II. (a) What are the different properties of moulding sand and write the different types of mould and green sand moulding? (7)
- (b) Define cores? Sketch and explain different types of cores. (8)
- OR**
- III. (a) Sketch and explain the cope and drag pattern and what are the different types of commonly used patterns and what are the allowances given to the patterns. (8)
- (b) Explain with neat sketches the gating designs and the gating design to prevent entering of impurities. (7)
- IV. (a) With neat sketches explain the injection moulding process, centrifugal casting process. (9)
- (b) Explain what are the inspections carried out in castings. (6)
- OR**
- V. (a) Discuss the different types of defects in castings and their origin. Sketch and explain any ten casting defects encountered in sand mould castings. (8)
- (b) Explain with neat sketches the die casting and continuous casting process. (7)
- VI. (a) Differentiate between the hot working and cold working process. (6)
- (b) With neat sketches explain the working principle of two, three, four and five high rolling mills. (9)
- OR**
- VII. (a) With neat sketches explain the following extrusion process:
 - (i) Backward extrusion (ii) Impact extrusion (iii) Hydrostatic extrusion
 - (iv) Closed cavity extrusion (8)
- (b) Explain with neat sketches the spinning operation carried out in sheet metal work and the process of coining and stretch forming. (7)
- VIII. With neat sketches explain the following welding process:
 - (i) TIG welding (ii) Metal inert gas welding (iii) Submerged arc welding (15)
- OR**
- IX. (a) What are the welding defects? Explain. (7)
- (b) With neat sketches explain the following welding process:
 - (i) Ultra sonic Welding (ii) Electron beam Welding (8)